Healthcare Big Data in Hong Kong: Development and implementation of artificial intelligence-enhanced predictive models for risk stratification

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Abstract

Routinely collected electronic health records (EHRs) data contain a vast amount of valuable information for conducting epidemiological studies. With the right tools, we can gain insights into disease processes and development, identify the best treatment and develop accurate models for predicting outcomes. Our recent systematic review has found that the number of big data studies from Hong Kong has rapidly increased since 2015, with an increasingly common application of artificial intelligence (AI). The advantages of big data are that i) the models developed are highly generalisable to the population, ii) multiple outcomes can be determined simultaneously, iii) ease of cross-validation by for model training, development and calibration, iv) huge numbers of useful variables can be analysed, v) static and dynamic variables can be analysed, vi) non-linear and latent interactions between variables can be captured, vii) artificial intelligence approaches can enhance the performance of prediction models. In this paper, we will provide several examples (cardiovascular disease, diabetes mellitus, Brugada syndrome, long QT syndrome) to illustrate efforts from a multi-disciplinary team to identify data from different modalities to develop models using territory-wide datasets, with the possibility of real-time risk updates by using new data captured from patients. The benefit is that only routinely collected data are required for developing highly accurate and high-performance models. AI-driven models outperform traditional models in terms of sensitivity, specificity, accuracy, area under the receiver operating characteristic and precision-recall curve, and F1 score. Web and/or mobile versions of the risk models allow clinicians to risk stratify patients quickly in clinical settings, thereby enabling clinical decision-making. Efforts are required to identify the best ways of implementing AI algorithms on the web and mobile apps.

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KW - Big Data

KW - Implementation

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